WEJ LAB

Evaluation Metrics

W. Evan Johnson, Ph.D.
Professor, Division of Infectious Disease
Director, Center for Data Science
Associate Director, Center for Biomedical Informatics and Health AI
Rutgers University – New Jersey Medical School

2026-10-14

RUTGERS HEALTH
Center for Data Science

Evaluation metrics

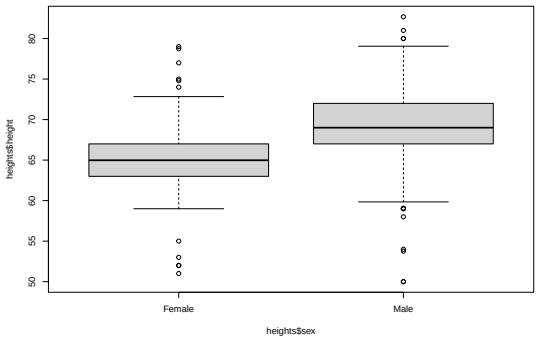
Before optimizing algorithms, we need to define what "better" means. We use `caret` and the `heights` data from dslabs.

```
library(tidyverse)
library(caret)
library(dslabs)
data(heights)
```

Evaluation metrics

Height by sex:

```
boxplot(heights$height~heights$sex)
```



Evaluation metrics

Simple example: predict sex from height.

```
y <- heights$sex
x <- heights$height
```

Evaluation metrics

- One predictor (height) and a categorical outcome (`Male`/`Female`)
- Heights overlap a lot, so we won't predict Y very accurately
- Can we beat guessing? We need a quantitative definition of "better"

Training and test sets

Algorithms are judged on new data. So we split the data:

- **Training set:** outcomes known; used to build the algorithm
- **Test set:** we pretend outcomes are unknown; used only to evaluate

WEJ LAB
RUTGERS HEALTH
Center for Data Science

Training and test sets

`caret::createDataPartition` randomly splits the data:

```
set.seed(2007)
test_index <- createDataPartition(y, times = 1,
                                  p = 0.5, list = FALSE)
```

`times`: number of splits; `p`: proportion in the index; `list`: return a list or not.

Training and test sets

Define the two sets:

```
test_set <- heights[test_index, ]
train_set <- heights[-test_index, ]
```

Develop using **only** the training set, then **freeze** the algorithm and evaluate it on the test set. The simplest metric is **overall accuracy**.

Overall accuracy

The simplest possible algorithm: guess.

```
y_hat <- sample(c("Male", "Female"),
                length(test_index), replace = TRUE)
```

This ignores height entirely.

Overall accuracy

ML functions in R usually expect categorical outcomes as factors:

```
y_hat <- sample(c("Male", "Female"),
                length(test_index), replace = TRUE) %>%
  factor(levels = levels(test_set$sex))
```

Overall accuracy

Overall accuracy is the proportion predicted correctly:

```
mean(y_hat == test_set$sex)
```

```
## [1] 0.5104762
```

About 50%. We are guessing!

Overall accuracy

Can we do better? On average, males are slightly taller:

```
heights %>% group_by(sex) %>%
  summarize(mean(height), sd(height))
```

```
## # A tibble: 2 x 3
##   sex    `mean(height)` `sd(height)`
##   <fct>           <dbl>        <dbl>
## 1 Female           64.9         3.76
## 2 Male             69.3         3.61
```

Overall accuracy

Another simple rule: predict `Male` if height is above 62 inches (about two SDs below the average male):

```
y_hat <- ifelse(x > 62, "Male", "Female") %>%
  factor(levels = levels(test_set$sex))
```

Accuracy goes from 0.50 to about 0.80:

```
mean(y == y_hat)
```

```
## [1] 0.7933333
```

Overall accuracy

Can we pick a better cutoff than 62? Try several and keep the best.

Optimize the cutoff on the training set only: the test set is only for evaluation.

Overall accuracy

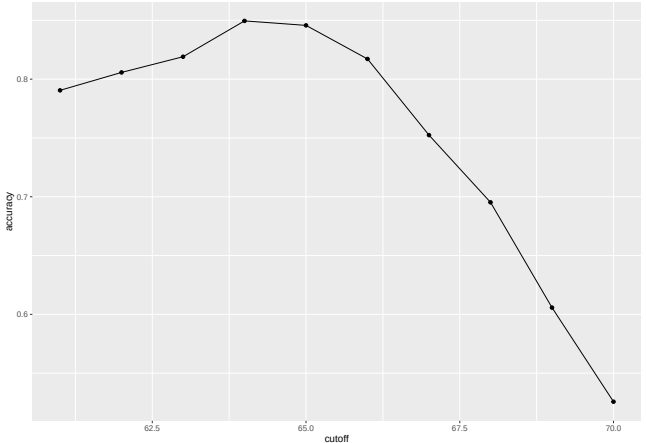
Training-set accuracy for 10 cutoffs:

```
cutoff <- seq(61, 70)
accuracy <- map_dbl(cutoff, function(x){
  y_hat <- ifelse(train_set$height > x, "Male", "Female") %>%
    factor(levels = levels(test_set$sex))
  mean(y_hat == train_set$sex)
})
```

WEJ LAB
RUTGERS HEALTH
Center for Data Science

Overall accuracy

Training-set accuracy by cutoff:



Overall accuracy

The maximum accuracy:

```
max(accuracy)
```

```
## [1] 0.8495238
```

and its cutoff:

```
best_cutoff <- cutoff[which.max(accuracy)]
best_cutoff
```

```
## [1] 64
```

Overall accuracy

Check this cutoff on the test set, so the estimate isn't over-optimistic:

```
y_hat <- ifelse(test_set$height >
  best_cutoff,"Male","Female") %>%
  factor(levels = levels(test_set$sex))
mean(y_hat == test_set$sex)
```

```
## [1] 0.8038095
```

The confusion matrix

Overall accuracy can be deceptive. The *confusion matrix* tabulates predictions against truth:

```
table(predicted = y_hat, actual = test_set$sex)
```

```
##           actual
## predicted Female Male
##    Female     48   32
##    Male       71  374
```

WEJ LAB
RUTGERS HEALTH
Center for Data Science

The confusion matrix

Accuracy computed separately by sex reveals a problem:

```
test_set %>%
  mutate(y_hat = y_hat) %>%
  group_by(sex) %>%
  summarize(accuracy = mean(y_hat == sex))
```

```
## # A tibble: 2 x 2
##   sex    accuracy
##   <fct>     <dbl>
## 1 Female    0.403
## 2 Male      0.921
```

The confusion matrix

Too many females are predicted male, because the **prevalence** of males is high:

```
prev <- mean(y == "Male")
prev
```

```
## [1] 0.7733333
```

The confusion matrix

In overall accuracy, the many errors for females are outweighed by correct calls for males. **This is a big problem in machine learning.**

Sensitivity and specificity

The four entries of the **confusion matrix**:

	Actually Positive	Actually Negative
Predicted positive	True positives (TP)	False positives (FP)
Predicted negative	False negatives (FN)	True negatives (TN)

Sensitivity and specificity

Sensitivity $= TP/(TP + FN)$: the proportion of actual positives called positive.

Also called the **true positive rate** (TPR) or **recall**.

Sensitivity and specificity

Specificity $= TN/(TN + FP)$: the proportion of actual negatives called negative.

Also called the **true negative rate** (TNR).

Sensitivity and specificity

Precision $= TP/(TP + FP)$: the proportion of predicted positives that are actually positive.

Also called **positive predictive value** (PPV). Unlike TPR and TNR, precision depends on prevalence: higher prevalence gives higher precision even when guessing.

Sensitivity and specificity

Summary of the terms:

Measure of	Name 1	Name 2	Definition	Probability representation
sensitivity	TPR	Recall	$\frac{TP}{TP+FN}$	$Pr(\hat{Y} = 1 \mid Y = 1)$
specificity	TNR	1-FPR	$\frac{TN}{TN+FP}$	$Pr(\hat{Y} = 0 \mid Y = 0)$
Precision	PPV		$\frac{TP}{TP+FP}$	$Pr(Y = 1 \mid \hat{Y} = 1)$

Here TPR is True Positive Rate, FPR is False Positive Rate, and PPV is Positive Predictive Value.

Sensitivity and specificity

`caret::confusionMatrix` computes all of these. It expects factors and treats the **first level** as positive ($Y = 1$); here that is `Female` (alphabetical).

```
cm <- confusionMatrix(data = y_hat,
                      reference = test_set$sex)
```

WEJ LAB
RUTGERS HEALTH
Center for Data Science

Sensitivity and specificity

Access the metrics directly:

```
cm$overall["Accuracy"]
```

```
##  Accuracy
## 0.8038095
```

```
cm$byClass[c("Sensitivity","Specificity", "Prevalence")]
```

```
## Sensitivity Specificity  Prevalence
##   0.4033613   0.9211823   0.2266667
```

WEJ LAB
RUTGERS HEALTH
Center for Data Science

Sensitivity and specificity

Overall accuracy is high despite low sensitivity, because female prevalence is low (0.23).

Missing females (low sensitivity) costs less accuracy than missing males would (low specificity).

Session Info

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.6.2
##
## Matrix products: default
## BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/libBLAS.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  LAPACK version 3.12.1
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/Denver
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] dslabs_0.9.1    caret_7.0-1     lattice_0.22-9  lubridate_1.9.5
##  [5] forcats_1.0.1   stringr_1.6.0   dplyr_1.2.1     purrr_1.2.2
##  [9] readr_2.2.0     tidyr_1.3.2     tibble_3.3.1    ggplot2_4.0.3
## [13] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] tidyselect_1.2.1     viridisLite_0.4.3    timeDate_4052.112
##  [4] farver_2.1.2         S7_0.2.2             fastmap_1.2.0
```

WEJ LAB

RUTGERS HEALTH
Center for Data Science